

# Support Vector Regression (1)

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- The SVM approach is valid for this task.
- The criterion is the same:
  - Find a set of constraints that define a set of loss or slack variables to define an empirical risk.
  - Minimize the empirical risk and the complexity in the same functional.

- The regression model is defined as

$$y_n = \mathbf{w}^\top \mathbf{x}_n + b + e_n$$

where  $y_n \in \mathbb{R}$  is the set of regressors (desired outputs), and  $e_n$  is the regression error for sample  $\mathbf{x}_n$

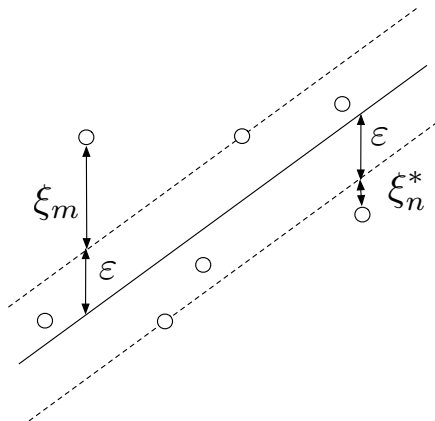
- The error is also continuous variable in  $\mathbb{R}$ , so the definition of positive slack variables must be defined as a function of the error norm

- In order to define positive slack variables, we consider the positive and negative error cases (Smola, 2003):

$$\begin{aligned} y_n - \mathbf{w}^\top \mathbf{x}_n - b &\leq \varepsilon + \xi_n \\ -y_n + \mathbf{w}^\top \mathbf{x}_n + b &\leq \varepsilon + \xi_n^* \\ \xi_n, \xi_n^* &\geq 0 \end{aligned}$$

- Interpretation:
  - We define an  $\varepsilon$  margin or  $\varepsilon$  tube as an error tolerance  $\pm\varepsilon$ .
  - If the error is less than  $|\varepsilon|$ , the slacks are dropped to zero.
  - Otherwise, the slacks are positive.
- We place as many samples as possible *inside* the margin while minimizing the loss on the samples *outside* the margin.

- The samples inside the margin have zero slack variables.



- The original development of the SVR minimizes the sum of the slack variables, which is a linear risk, plus the complexity.

$$\begin{aligned} & \text{Minimize } \|\mathbf{w}\|^2 + C \sum_{n=1}^N (\xi_n + \xi_n^*) \\ & \text{subject to } \begin{cases} y_n - \mathbf{w}^\top \mathbf{x}_n - b - \varepsilon - \xi_n \leq 0 \\ -y_n + \mathbf{w}^\top \mathbf{x}_n + b - \varepsilon - \xi_n^* \leq 0 \\ \xi_n, \xi_n^* \geq 0 \end{cases} \end{aligned}$$

- The Lagrange optimization involves four sets of positive Lagrange multipliers:  $\alpha_n, \alpha_n^*$  and  $\mu_n, \mu_n^*$ .
- We must *maximize* the two first constraints, minimize the other two. The Lagrangian is

$$\begin{aligned} L_L = & ||\mathbf{w}||^2 + C \sum_{n=1}^N (\xi_i + \xi_i^*) \\ & - \sum_{n=1}^N \alpha_i (-y_n + \mathbf{w}^\top \mathbf{x}_n + b + \varepsilon + \xi_n) \\ & - \sum_{n=1}^N \alpha_i^* (y_n - \mathbf{w}^\top \mathbf{x}_n - b + \varepsilon + \xi_n^*) \\ & - \sum_{n=1}^N (\mu_n \xi_n + \mu_n^* \xi_n^*) \end{aligned}$$

- A procedure similar to the one used for the SVC, that takes into account the KKT complementary conditions for the product constraint-Lagrange multipliers, leads to the minimization of

$$L_d = -\frac{1}{2}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*)^\top \mathbf{K}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) + (\boldsymbol{\alpha} - \boldsymbol{\alpha}^*)^\top \mathbf{y} - \varepsilon \mathbf{1}^\top (\boldsymbol{\alpha} + \boldsymbol{\alpha}^*)$$

with  $0 \leq \alpha_n, \alpha_n^* \leq C$

- This is a quadratic form that has the property of existence and uniqueness of a solution.
- The optimization of this functional is almost exactly equal to the one needed for the SVC.

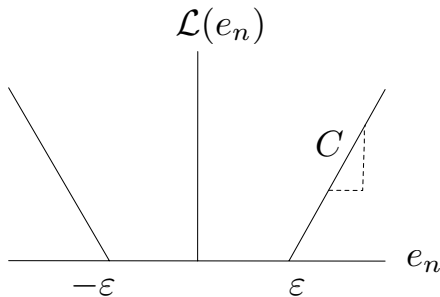


- The set of parameters is a function of the data

$$\mathbf{w} = \sum_{n=1}^N (\alpha_n - \alpha_n^*) \mathbf{x}_n$$

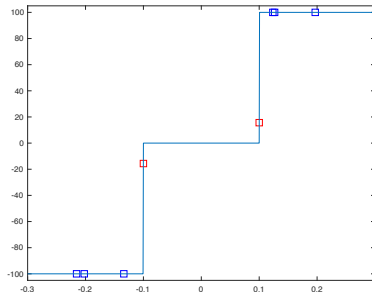
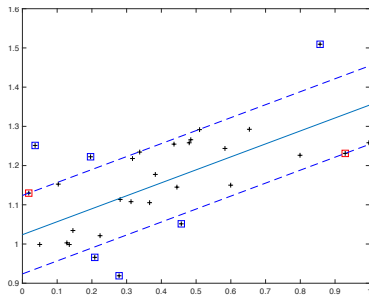
- The bias is obtained the same way as the one of the SVC, using the KKT conditions.
- The samples inside the margin have null dual variables  $\alpha_n$ . The ones on the margin or outside have positive values in their dual variables  $\alpha_n$ .

- The implicit cost or loss function over the errors is linear outside the margin.



- It can be shown that the values of the dual variables are the derivative of the cost function.

# Example of a linear SVR



- The application of the SVM criterion to regression through a reinterpretation of the margin.
- The practical optimization of the SVR.
- The properties of the dual variables in relation to their corresponding support vectors.